Trodimenzijske kontinuirane slučajne varijable – nezavisne

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4. prosinca 2024.

1 Funkcija gustoće (vjerojatnosti)

$$f(x,y,z) = 8xyz, \qquad 0 \leq x,y,z \leq 1 \qquad (0 \text{ inače})$$

Provjera normiranosti:

$$\int_0^1 \int_0^1 \int_0^1 8xyz\,dx\,dy\,dz = 8 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = 1 \quad \checkmark$$

2 Marginalne funkcije gustoće

1. primjer

$$f(x) = \int_0^1 \int_0^1 8xyz\,dy\,dz = 2x$$

2. primjer

$$f(y) = \int_0^1 \int_0^1 8xyz\,dx\,dz = 2y$$

3. primjer

$$f(z) = \int_0^1 \int_0^1 8xyz\,dx\,dy = 2z$$

$$f(x)f(y)f(z) = 8xyz = f(x,y,z) \longrightarrow \text{nezavisne varijable}$$

Dvodimenzijske marginalne gustoće:

$$f(x,y) = \int_0^1 8xyz\,dz = 4xy = f(x)f(y) \longrightarrow \text{nezavisne varijable} \tag{1}$$

$$f(x,z) = \int_0^1 8xyz\,dy = 4xz = f(x)f(z) \longrightarrow \text{nezavisne varijable} \tag{2}$$

$$f(y,z) = \int_0^1 8xyz\,dx = 4yz = f(y)f(z) \longrightarrow \text{nezavisne varijable} \tag{3}$$

3 Uvjetne funkcije gustoće

Opća formula: uvjetna gustoća dobiva se dijeljenjem zajedničke gustoće *zajedničkom* gustoćom uvjetnih varijabli, npr. $f(x \mid y,z) = \dfrac{f(x,y,z)}{f(y,z)}$. (Zbog nezavisnosti ovdje vrijedi $f(y,z) = f(y)f(z)$, pa se nazivnik smije rastaviti na produkt marginalnih gustoća.)

$$f(x \mid y,z) = \frac{f(x,y,z)}{f(y,z)} = \frac{8xyz}{4yz} = 2x = f(x) \longrightarrow \text{nezavisne varijable} \tag{4}$$

$$f(y \mid x,z) = \frac{f(x,y,z)}{f(x,z)} = \frac{8xyz}{4xz} = 2y = f(y) \tag{5}$$

$$f(z \mid x,y) = \frac{f(x,y,z)}{f(x,y)} = \frac{8xyz}{4xy} = 2z = f(z) \tag{6}$$

$$f(x,y \mid z) = \frac{f(x,y,z)}{f(z)} = \frac{8xyz}{2z} = 4xy = f(x)f(y) \longrightarrow \text{nezavisne varijable} \tag{7}$$

$$f(x,z \mid y) = \frac{f(x,y,z)}{f(y)} = \frac{8xyz}{2y} = 4xz = f(x)f(z) \tag{8}$$

$$f(y,z \mid x) = \frac{f(x,y,z)}{f(x)} = \frac{8xyz}{2x} = 4yz = f(y)f(z) \tag{9}$$

4 Matematička očekivanja i varijance

$$E(X) = \int_a^b x f(x)\,dx$$

ili, budući da je $f(x) = \int_0^1 \int_0^1 8xyz\,dy\,dz$,

$$E(X) = \int_0^1 \int_0^1 \int_0^1 x\,(8xyz)\,dx\,dy\,dz = \frac{2}{3} \tag{10}$$

$$E(Y) = \int_0^1 y \cdot 2y\,dy = \frac{2}{3} \tag{11}$$

$$E(Z) = \int_0^1 z \cdot 2z\,dz = \frac{2}{3} \tag{12}$$

$$V(X) = \int_a^b x^2 f(x)\,dx - E(X)^2 = \int_0^1 x^2 \cdot 2x\,dx - \left(\frac{2}{3}\right)^2 = \frac{1}{18} \tag{13}$$

$$V(Y) = \int_0^1 y^2 \cdot 2y\,dy - \left(\frac{2}{3}\right)^2 = \frac{1}{18} \tag{14}$$

$$V(Z) = \int_0^1 z^2 \cdot 2z\,dz - \left(\frac{2}{3}\right)^2 = \frac{1}{18} \tag{15}$$

5 Uvjetno očekivanje, kovarijance i koeficijenti korelacije

5.1 Uvjetno očekivanje

$$E(X \mid y,z) = \int_a^b x f(x \mid y,z)\,dx = \int_a^b x f(x)\,dx = E(X) \longrightarrow \text{nezavisne varijable} \tag{16}$$

$$E(Y \mid x,z) = E(Y), \qquad E(Z \mid x,y) = E(Z)$$

5.2 Kovarijance

$$\operatorname{Cov}(X,Y) = \int_a^b \int_c^d xy\, f(x,y)\, dx\, dy - E(X)E(Y) \tag{17}$$

$$= \int_0^1 \int_0^1 xy \cdot 4xy\, dx\, dy - E(X)E(Y) = \frac{4}{9} - \frac{2}{3}\cdot\frac{2}{3} = 0 \tag{18}$$

ili

$$\operatorname{Cov}(X,Y) = \int_a^b \int_c^d \int_e^f \big(x - E(X)\big)\big(y - E(Y)\big) f(x,y,z)\, dx\, dy\, dz \tag{19}$$

$$= \int_0^1 \int_0^1 \int_0^1 \left(x - \frac{2}{3}\right)\left(y - \frac{2}{3}\right)(8xyz)\, dx\, dy\, dz = 0 \tag{20}$$

ili

$$\operatorname{Cov}(X,Y) = E(XY) - E(X)E(Y) \tag{21}$$

$$E(XY) = \int_0^1 \int_0^1 \int_0^1 xy\,(8xyz)\, dx\, dy\, dz = \frac{4}{9} \tag{22}$$

$$\operatorname{Cov}(X,Y) = \frac{4}{9} - \frac{2}{3}\cdot\frac{2}{3} = 0 \tag{23}$$

$$\operatorname{Cov}(X,Z) = 0, \qquad \operatorname{Cov}(Y,Z) = 0$$

5.3 Koeficijenti korelacije

$$\rho(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\sigma(X)\,\sigma(Y)} = 0, \qquad \rho(X,Z) = \frac{\operatorname{Cov}(X,Z)}{\sigma(X)\,\sigma(Z)} = 0, \qquad \rho(Y,Z) = \frac{\operatorname{Cov}(Y,Z)}{\sigma(Y)\,\sigma(Z)} = 0$$